

GROSSEVICH, Russian Federation

WMO: 318230

Lat: 47.971N

Lon: 139.535E

Elev: 3

StdP: 101.29

Time Zone: 10.00 (E10)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -22.6	(c) -20.9	(d) -29.6	(e) 0.2	(f) -20.2	(g) -27.9	(h) 0.3	(i) -19.0	(j) 15.5	(k) -10.4	(l) 13.4	(m) -9.5	(n) 5.2	(o) 300	(p) 0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.2	(c) 20.4	(d) 17.1	(e) 18.9	(f) 16.7	(g) 17.9	(h) 16.2	(i) 18.0	(j) 19.3	(k) 17.2	(l) 18.4	(m) 16.5	(n) 17.6	(o) 3.9	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	12.6	18.5	16.7	11.9	17.8	16.0	11.4	17.2	51.2	19.5	48.7	18.6	46.5	17.6	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.4	11.6	9.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-25.2	20.3	1.3	1.5	-26.1	21.3	-26.9	22.2	-27.6	23.0	-28.6	24.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-13.8	-10.5	-5.1	0.6	4.5	8.4	13.1	16.0	12.6	5.9	-3.4	-10.3
	DBStd	10.20	3.81	3.97	3.21	2.11	2.36	2.13	1.96	1.95	2.53	3.39	4.76	4.27
	HDD10.0	3456	737	573	467	282	171	56	2	0	7	133	402	628
	HDD18.3	6119	995	807	726	532	428	297	162	78	172	387	652	886
	CDD10.0	384	0	0	0	0	1	9	98	186	86	4	0	0
	CDD18.3	6	0	0	0	0	0	0	1	5	0	0	0	0
	CDH23.3	10	0	0	0	0	0	0	1	9	1	0	0	0
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	4.4	5.0	5.0	4.7	4.2	3.6	3.4	3.2	3.7	4.8	5.2	5.2	5.6
Precipitation	PrecAvg	978	41	34	60	65	84	96	129	145	102	87	76	60
	PrecMax	1237	86	110	123	112	145	188	213	235	183	200	150	176
	PrecMin	692	10	3	21	20	31	26	35	68	39	21	20	15
	PrecStd	146	21	22	32	25	27	36	50	45	40	45	33	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.0	0.7	4.5	10.1	15.6	16.7	20.6	24.3	20.8	15.7	8.1	1.7
		MCWB	-2.2	-1.2	1.0	4.0	7.9	13.1	16.8	17.7	15.6	11.4	5.9	0.1
	2%	DB	-3.4	-1.3	1.7	6.7	11.6	14.1	17.6	21.3	18.9	13.9	6.1	-0.5
		MCWB	-4.8	-3.1	-0.9	2.6	7.2	12.0	16.2	18.0	16.0	10.9	3.7	-2.1
	5%	DB	-5.8	-2.4	0.3	4.8	9.1	12.7	16.6	19.8	17.8	12.7	4.8	-2.3
		MCWB	-7.2	-3.9	-1.7	2.0	6.4	11.1	15.5	17.3	15.3	10.3	2.6	-4.2
	10%	DB	-7.9	-4.0	-0.6	3.6	7.7	11.5	15.7	18.7	16.8	11.5	3.3	-4.1
		MCWB	-9.2	-5.9	-2.3	1.8	5.7	10.4	15.0	16.9	14.6	9.3	1.0	-5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.4	-0.9	1.5	5.2	9.3	13.8	18.0	20.0	18.0	13.2	6.2	0.0
		MCDB	-1.3	0.8	3.4	7.8	12.5	15.7	18.9	21.9	19.2	14.5	7.2	0.9
	2%	WB	-4.9	-2.7	-0.1	3.4	7.8	12.4	16.5	18.5	16.7	11.9	4.4	-1.8
		MCDB	-3.6	-1.4	1.1	5.0	9.9	13.7	17.3	20.0	18.4	13.1	5.6	-0.4
	5%	WB	-7.3	-3.9	-1.1	2.7	6.7	11.3	15.7	17.8	15.8	10.8	2.8	-4.1
		MCDB	-6.0	-2.4	-0.2	4.0	8.4	12.4	16.5	19.1	17.2	12.1	4.4	-2.4
	10%	WB	-9.2	-5.8	-1.9	2.1	6.0	10.5	15.0	17.2	15.1	9.6	1.2	-5.7
		MCDB	-7.9	-4.2	-0.9	3.3	7.2	11.3	15.6	18.4	16.5	11.3	3.1	-4.2
Mean Daily Temperature Range	5% DB	MDBR	8.3	8.2	6.6	4.5	4.2	3.6	3.2	4.2	6.4	7.7	7.9	7.8
		MCDBR	9.2	7.9	7.3	7.2	7.8	6.4	4.8	6.6	7.7	8.5	8.2	9.0
		MCWBR	8.2	6.8	5.9	4.9	4.6	4.5	3.6	4.3	5.6	6.9	7.1	7.7
	5% WB	MCDBR	8.9	7.5	6.2	5.8	6.1	5.4	4.4	5.4	5.9	7.5	7.4	8.2
		MCWBR	8.0	6.6	5.6	4.6	4.3	4.2	3.4	4.1	4.9	6.8	7.0	7.2
Clear-Sky Solar Irradiance	taub		0.298	0.339	0.409	0.500	0.537	0.512	0.477	0.439	0.396	0.398	0.352	0.309
	taud		2.213	2.088	1.953	1.834	1.843	2.020	2.187	2.308	2.347	2.167	2.168	2.175
	Ebn at Noon		756	792	785	750	740	765	788	804	805	718	669	679
	Edh at Noon		80	114	154	192	198	167	140	119	104	105	84	74
All-Sky Solar Radiation	RadAvg		1.68	2.76	3.88	4.50	4.87	5.01	4.71	4.32	3.74	2.57	1.68	1.31
	RadStd		0.13	0.14	0.33	0.49	0.37	0.66	0.48	0.37	0.34	0.21	0.15	0.10

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page